Abdur Rahman Shaik

AI Engineer | LLM, Agentic AI & Full-Stack

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LinkedIn | GitHub | Portfolio

AI engineer with 5+ years building, deploying, and supporting production applications, APIs, and AI/GenAI systems. Specializes in LLM and agentic applications, RAG pipelines, and multi-agent orchestration, with strong backend engineering in Python and Java, REST APIs, and cloud-native deployment on Azure and AWS. Ships end to end with Docker, Kubernetes, and CI/CD, and thrives on translating complex problems into reliable, evaluated, compliant solutions. Master of Engineering in Computer Science (GPA 3.92/4.0).

TECHNICAL SKILLS

AI & LLMs: LLMs, Generative AI, agentic AI, multi-agent orchestration (LangGraph, CrewAI), LangChain, RAG, vector databases (Pinecone), embeddings, prompt engineering, evaluation (RAGAS, LangSmith), Azure OpenAI, LLM integration (Anthropic, OpenAI, Groq)
ML & CV: PyTorch, TensorFlow / Keras, scikit-learn, computer vision (OpenCV), applied machine learning, model training and evaluation
Languages: Python, Java, JavaScript, TypeScript, SQL, C (coursework)
APIs & Backend: REST API design, FastAPI, Django, Spring Boot, Node.js, microservices, event-driven architecture (Kafka), enterprise integrations
Cloud & DevOps: Azure (Azure OpenAI, AKS, Azure DevOps, VMs, Blob Storage), AWS (S3, Lambda), Docker, Kubernetes, CI/CD (Azure DevOps, Jenkins, Argo CD, GitHub Actions), Git
Data: PostgreSQL, MySQL, MongoDB, Snowflake, SQL and NoSQL, data pipelines, pandas
Practices: Agile and Scrum, automated testing, observability (OpenTelemetry, Grafana, LangSmith), guardrails and structured-output validation, documentation

PROFESSIONAL EXPERIENCE

AI Software Engineer | Jabil Circuit Inc. Florence, KY | July 2026 to Present

AI Applications & Systems Testing

- Build AI-driven software and data workflows in Python, applying LLM, RAG, and computer-vision techniques to manufacturing quality and defect-detection use cases.
- Engineered a computer-vision inspection system with Python, PyTorch, and OpenCV to detect PCB defects (missing components, solder bridges, misplacements), reducing manual inspection time by 38% during pilot deployment.
- Built an AI quality-control copilot using LangChain, Pinecone, and Azure OpenAI, integrating RAG over manufacturing SOPs and defect reports to cut engineer information-retrieval time by 45%.
- Developed an asynchronous FastAPI inference service and containerized AI workloads with Docker, deploying object-detection models on edge infrastructure and improving defect-detection recall by 14 percentage points.
- Test and validate enterprise server and system software in a production environment, documenting results for reliability and traceability.
- Collaborated with engineering and operations teams on evaluation, tuning, and integration of AI capabilities into production workflows.

AI Engineer | Optum (UnitedHealth Group) Remote, USA | August 2025 to April 2026

Agentic AI & Clinical Automation

- Engineered a multi-agent orchestration workflow using LangGraph and CrewAI to automate prior-authorization document review, cutting manual review time by 42%.
- Built a RAG pipeline with LangChain and Pinecone over clinical policy documents to power a natural-language Q&A copilot used daily by care coordinators.
- Integrated Azure OpenAI GPT models into agent workflows and validated output quality with RAGAS and prompt-safety testing to meet clinical compliance standards.
- Added guardrails and structured-output validation to agent responses, reducing hallucination-related escalations from downstream teams.
- Developed FastAPI microservices for agent endpoints and deployed containerized workloads on Azure Kubernetes Service (AKS) using Docker.

- Set up CI/CD pipelines in Azure DevOps, enabling automated testing and promotion of agent services across dev, staging, and production.
- Implemented LangSmith tracing to debug multi-agent execution paths, cutting the time needed to root-cause production incidents.
- Partnered with clinical SMEs to define agent evaluation rubrics, improving response accuracy by 18% over baseline.
- Automated recurring intake and document-extraction workflows using n8n and MCP-based tool integrations, reducing manual data-entry effort by 30% for the operations team.

Full-Stack Software Engineer | Phenom Hyderabad, India | February 2022 to July 2024

Backend, Data & AI Features

- Architected an event-driven logging and analytics platform with Apache Kafka, Spring Boot, Loki, AWS S3, and REST APIs, processing 1M+ events/day and improving log-processing scalability by 35%.
- Built and embedded a LangChain-based GenAI chatbot with prompt engineering and platform-data integration, enabling 100+ users to query analytics in natural language and reducing manual data-lookup time by 45%.
- Co-developed a PyTorch user-behavior model and integrated inference into the full-stack application, improving behavioral prediction accuracy by 18%.
- Designed and built microservices and REST APIs in Java and Spring Boot for high-traffic enterprise clients including MERCK and Regions Bank.
- Supported production deployments with Docker, Jenkins, Argo CD, OpenTelemetry, and Grafana, instrumenting services for observability.
- Integrated relational and NoSQL databases (PostgreSQL, MongoDB, Snowflake) and optimized queries for performance.
- Collaborated in Agile ceremonies with product and cross-functional teams, contributing to code reviews and technical documentation.

Software Engineer | Integr8 Software Products & Services Hyderabad, India | May 2020 to December 2021

Full-Stack & Automation

- Engineered TestVault, a test reporting and automation platform using Python, Django, pandas, Java, and Spring Boot, reducing manual testing effort by 35% across 5+ projects and improving reporting accuracy.
- Developed backend services, REST APIs, and full-stack interfaces in Python, Java, React, and TypeScript, reducing API response latency by 30% through query optimization and backend improvements.
- Built a shared React component library and integrated SQL-backed services with Docker and CI/CD, reducing duplicate UI code by 41% and accelerating feature development by 25%.
- Designed relational and NoSQL data models and integrated services across a microservices architecture.
- Diagnosed and resolved production issues across the stack, improving reliability and maintainability.
- Collaborated in Agile teams on design, code reviews, testing, and mentoring of junior developers.

PROJECTS

Aria, Personal AI Assistant | *Python, FastAPI, MongoDB, React, Multi-LLM* | Portfolio

- Built a multi-provider virtual assistant integrating 4 LLM providers with fallback routing, natural-language question answering, and MongoDB persistence, reducing repeated integration code by 30% via reusable abstractions.

Flight Search AI Agent | *Python, Agno / PhiData, Groq, SerpApi, Streamlit* | Live Demo

- Built an autonomous LLM agent orchestrating tool and function calls to retrieve and summarize live data from natural-language queries, reducing manual search steps by 60% in test scenarios.

AI-Powered Automated Code Review System | *Python, FastAPI, LLM APIs, GitHub Webhooks* | Link

- Built a full-stack platform with a Python/FastAPI REST backend and GitHub webhooks that analyzes pull requests and returns structured developer feedback via reusable prompt templates.

EDUCATION

Master of Engineering, Computer Science | University of Cincinnati | 2026 | GPA 3.92/4.0

Bachelor of Technology, Electronics and Communication Engineering | Sreenidhi Institute of Science and Technology | 2022